

Practice Test 2

Module 1

1.

$$\begin{aligned}y &> -3x + 5 \\ y &> 2x\end{aligned}$$

Which ordered pair (x, y) is a solution to the given system of inequalities?



2. If $3x - y = 11$ and $2x - 2y = 2$, what is the value of $x + y$?



[PREV. QUESTION](#)[NEXT QUESTION](#)

☒ A

(1, 4)

90.0%

☐ B

(2, 1)

<5.0%

☐ C

(2, 4)

<5.0%

☐ D

(4, 3)

<5.0%

[COME BACK LATER](#)

[PREV. QUESTION](#)[NEXT QUESTION](#)

☐ A

3

<5.0%

☐ B

5

5.2%

☒ C

9

91.6%

☐ D

13

<5.0%

[COME BACK LATER](#)

3. If a is a solution to the equation $|2x - 3| = 11$ and $a > 0$, what is the value of a ?

PREV. QUESTION

NEXT QUESTION

7

Correct Answer:

7

COME BACK LATER

4. $V(t) = 25(0.88)^{\frac{t}{3}}$
The given equation models the value that remains, $V(t)$, in millions of dollars, of a new piece of equipment after t years. What is the meaning of 25 in this context?

PREV. QUESTION

NEXT QUESTION

- ☒ A The number of years until the piece of equipment has no value.
<5.0%
- ☐ B The value of the equipment, in millions of dollars, after the first year.
7.6%
- ☐ C The number of years until the company expects to replace this piece of equipment.
<5.0%
- ☒ D The value of equipment, in millions of dollars, when new.
86.7%

5. The radius of a right cylinder is 6 inches, and its height is twice the length of the radius. What is the volume of the cylinder, in cubic inches?

PREV. QUESTION

NEXT QUESTION

- ☐ A 108π
<5.0%
- ☐ B 216π
<5.0%
- ☒ C 432π
92.7%
- ☐ D 864π
<5.0%

COME BACK LATER

6. The table shows the average weekly bread sales at Max's Bakery.

	Sliced	Unsliced	Total
Rye	54	41	95
Whole Grain	119	84	203
Sourdough	32	55	87
Total	205	180	385

If a customer who purchased unsliced bread is selected at random, what is the probability that the customer has purchased whole grain bread?



PREV. QUESTION
NEXT QUESTION

A
 $\frac{84}{203}$
<5.0%

B
 $\frac{84}{180}$
90.6%

C
 $\frac{203}{385}$
<5.0%

D
 $\frac{180}{203}$
<5.0%

COME BACK LATER

7. A group of friends is eating at a diner. Each entree costs \$5.25, and each soda costs \$1.50. If d people order an entree and p people order a soda, which of the following expressions represents the total number of dollars paid by the group?



PREV. QUESTION
NEXT QUESTION

A
 $5.25d - 1.5p$
<5.0%

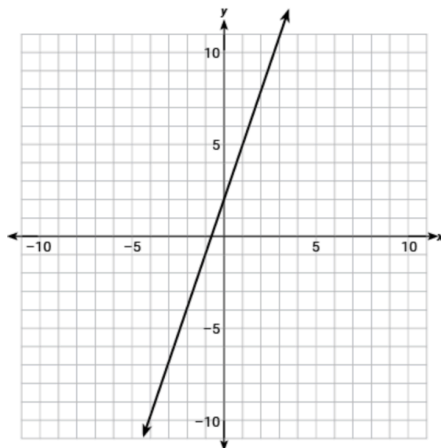
B
 $5.25d + 1.5p$
97.6%

C
 $5.25p + 1.5d$
<5.0%

D
 $6.75dp$
<5.0%

COME BACK LATER

8.



The graph of linear function $g(x)$ is shown above. What is the value of $g(2)$?

PREV. QUESTION

NEXT QUESTION

☐ A 0
5.4%

☐ B 5
<5.0%

☒ C 8
88.9%

☐ D 10
<5.0%

COME BACK LATER

9.

$$kx^2 + 98 = 1,122$$

If the solution to the given equation is a positive whole number, what could be a value of k ?

PREV. QUESTION

NEXT QUESTION

☐ A 2
<5.0%

☒ B 4
77.1%

☐ C 8
6.6%

☐ D 32
13.4%

COME BACK LATER

10. The expression $\frac{x^2+6x+8}{x^2+2x-8}$ is equivalent to which expression?

PREV. QUESTION

NEXT QUESTION

✕ **A** $\frac{x+2}{x-2}$
85.3%

✕ **B** $\frac{x+8}{x-1}$
<5.0%

✕ **C** $\frac{3x+4}{x-4}$
11.5%

✕ **D** 6
<5.0%

🚩 COME BACK LATER

11. $\frac{\sqrt{x^2-9}}{a} = 1$

In the given equation, $x < 0$ and $a = 4$. What is the value of $x - 5$?

PREV. QUESTION

NEXT QUESTION

✕ **A** -13
<5.0%

✕ **B** -10
77.6%

✕ **C** -5
15.5%

✕ **D** -3
<5.0%

🚩 COME BACK LATER

12. $y = 3x + 2$
 $5y = 13x - 14$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

PREV. QUESTION

NEXT QUESTION

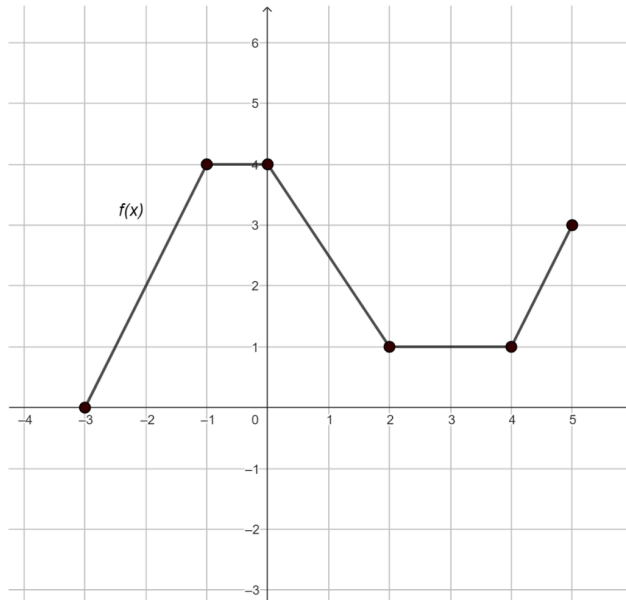
22

Correct Answer:

22

🚩 COME BACK LATER

13.



The complete graph of the function f in the xy -plane is shown. For how many values of x is $f(x)$ equal to 2.5?

PREV. QUESTION NEXT QUESTION

✕ A 0
6.8%

✕ B 1
14.9%

✕ C 2
11.9%

✕ D 3
66.4%

COME BACK LATER

14.

$$\begin{aligned}(t - 5) + 2(s - 3) &= 8 \\ 2(t - 5) - 3(s - 3) &= -19\end{aligned}$$

The solution to the given system of equations is (t, s) . What is the value of s ?

PREV. QUESTION NEXT QUESTION

8

Correct Answer:

8

COME BACK LATER

15.

$$(5x^4 + x^3) - (3x^3 + 4x^2) - (x^2 - 8x)$$

The given expression can be written in the form $ax^4 + bx^3 + cx^2 + dx$, where a , b , c , and d are constants. What is the value of $c + d$?

PREV. QUESTION NEXT QUESTION

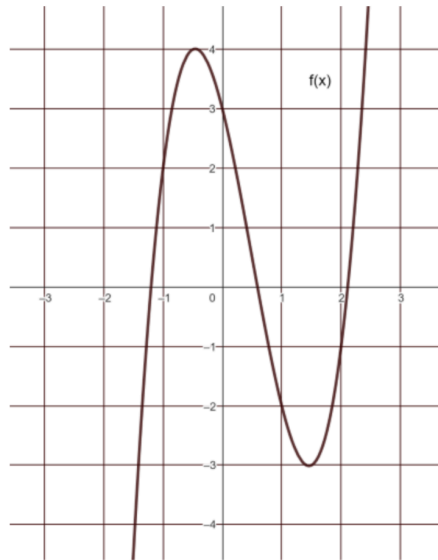
3

Correct Answer:

3

COME BACK LATER

16.



The graph of $f(x)$ is shown. If $g(x) = f(x) + 2$, what must be true of $g(x)$?

17. What is the value when 80 is increased by 200%?

PREV. QUESTION

NEXT QUESTION

✕ ☐ A $g(-2) = 3$ and $g(-1) = -2$
8.3%

✕ ☐ B $g(0) = 1$ and $g(1) = -4$
9.6%

✕ ☒ C $g(0) = 5$ and $g(1) = 0$
74.9%

✕ ☐ D $g(2) = 3$ and $g(3) = -2$
7.2%

🚩 COME BACK LATER

PREV. QUESTION

NEXT QUESTION

✕ ☐ A 160
24.7%

✕ ☐ B 200
<5.0%

✕ ☒ C 240
74.1%

✕ ☐ D 280
<5.0%

🚩 COME BACK LATER

18. ABC is a right triangle, where angle B is 90° and angle C is 30° . If $AC = 32$ inches, what is the area of triangle ABC , in inches?

PREV. QUESTION

NEXT QUESTION

☒ A $16\sqrt{2}$
<5.0%

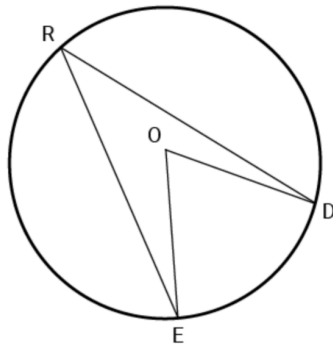
☒ B $16\sqrt{3}$
12.9%

☒ C $128\sqrt{2}$
8.5%

☒ D $128\sqrt{3}$
74.7%

COME BACK LATER

19.



The circle shown has center O . If the measure of angle DRE is 35° , what is the measure of angle EOD ?

PREV. QUESTION

NEXT QUESTION

☒ A 35°
15.5%

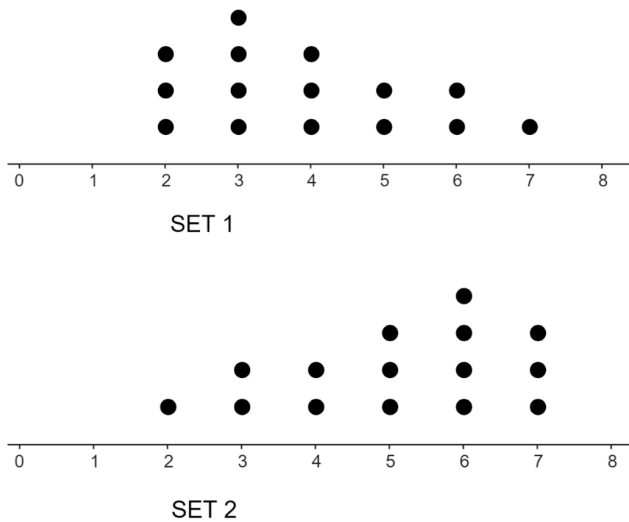
☒ B 55°
13.2%

☒ C 60°
7.8%

☒ D 70°
63.5%

COME BACK LATER

20.



The dot plots shown summarize the data in two sets, Set 1 and Set 2. Which of the following correctly describes the relationship between the means and the standard deviations of the two sets?

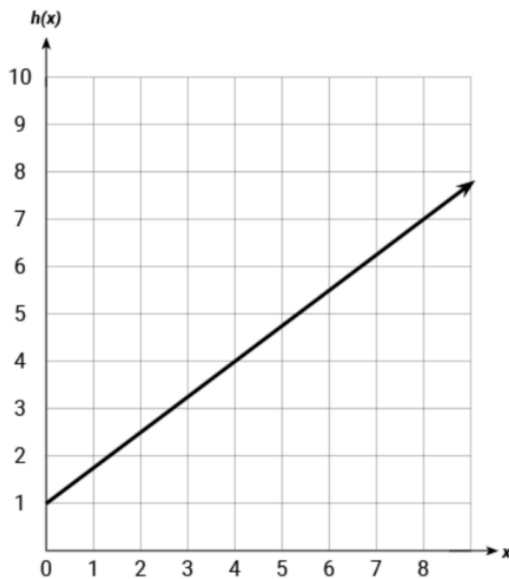
PREV. QUESTION

NEXT QUESTION

- ☒ A 11.9% The standard deviation of Set 1 is greater than the standard deviation of Set 2, and the mean of Set 1 is less than the mean of Set 2.
- ☒ B 7.6% The standard deviation of Set 1 is less than the standard deviation of Set 2, and the mean of Set 1 is greater than the mean of Set 2.
- ☒ C 75.1% The standard deviation of the two sets is equal, and the mean of Set 1 is less than the mean of Set 2.
- ☒ D 5.4% The standard deviation of the two sets is equal, and the mean of Set 1 is greater than the mean of Set 2.

COME BACK LATER

21.



The graph of the function $h(x)$ shows how high, in meters, a balloon is in the air after x seconds. What is the best interpretation of the function $h(x) + 3$?

PREV. QUESTION

NEXT QUESTION

- ☒ A <5.0% The balloon is released at the same height and the height of the balloon is tripled each second.
- ☒ B 30.9% The balloon is released at a height of 3 meters and the height of the balloon increases at the same rate.
- ☒ C 6.1% The balloon is released 3 seconds later and the height of the balloon increases by 3 meters each second.
- ☒ D 59.6% The balloon is released at a height of 4 meters and the height of the balloon increases at the same rate.

COME BACK LATER

22. What is the measure of an exterior angle of a regular hexagon, in degrees?

[PREV. QUESTION](#)[NEXT QUESTION](#)

☒ A

30
5.1%

☐ B

45
5.0%☒ C☐ D[COME BACK LATER](#)

Module 2

1. Maximum Mail Order Company upgrades its computers each year. The function $P(t) = 36t + 410$ models the company's budget for upgrades, in thousands of dollars, t years after 2004. What is the best interpretation of 36 in this context?

[PREV. QUESTION](#)[NEXT QUESTION](#)

☐ A

The number of years until the budget is completely spent
<5.0%

☐ B

The amount budgeted, in thousands of dollars, in the year 2004
7.5%

☒ C

The annual increase, in thousands of dollars, in the budget each year after 2004
89.3%

☐ D

The number of years in which the budget will increase after 2004
<5.0%

[COME BACK LATER](#)

2. Which expression is equivalent to $\frac{4x^2-3}{2x+\sqrt{3}}$?

× ☐ A $2x - 3$
<5.0%

× ☒ B $2x - \sqrt{3}$
90.5%

× ☐ C $2x + \sqrt{3}$
<5.0%

× ☐ D $2x + 3$
<5.0%

PREV. QUESTION

NEXT QUESTION

COME BACK LATER

3. $y - 7 = 3^x - 5$

What is the y -intercept for the given equation?

PREV. QUESTION

NEXT QUESTION

3

Correct Answer:

3

COME BACK LATER

4. A linear model estimates that the mean weight of a calf in New Zealand, at birth, is 34 kilograms. At 180 days after birth, the mean weight is 272 kilograms. Which function best defines this model, where $f(x)$ is the mean weight of a calf, in kilograms, x days after it is born, for $0 \leq x \leq 180$?

× ☒ A $f(x) = 0.76x + 34$
<5.0%

× ☒ B $f(x) = 0.76x + 180$
<5.0%

× ☐ C $f(x) = 1.32x + 34$
91.5%

× ☒ D $f(x) = 34x + 272$
<5.0%

PREV. QUESTION

NEXT QUESTION

COME BACK LATER

5.

x	$f(x)$
1	3
3	6
5	9
7	12
9	15

PREV. QUESTION

NEXT QUESTION

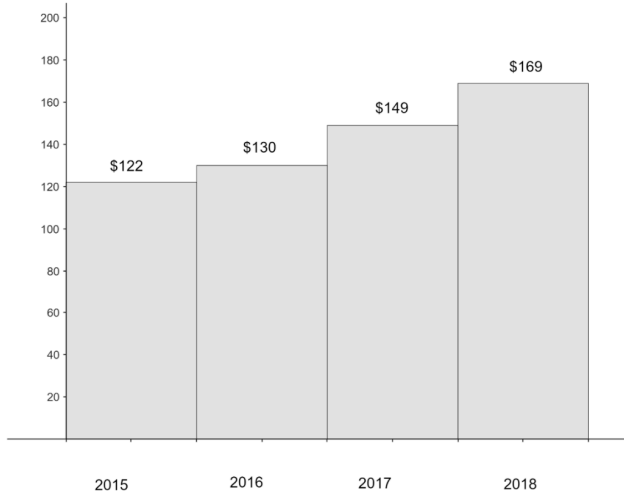
Correct Answer:

15

COME BACK LATER

For the function f , the table shows some values of x and their corresponding values of $f(x)$. If $f(1) = t$, what is the value of $f(3t)$?

6.



The chart shows the average cost of housing, in dollars per square foot, in Nebraska from 2015 to 2018. By approximately what percent did the average cost of housing increase over this period?

PREV. QUESTION

NEXT QUESTION

A

20%

<5.0%

B

28%

10.0%

C

39%

84.8%

D

47%

<5.0%

COME BACK LATER

7.

$$p(t) = 2,034(3)^{\frac{t}{14}}$$

The given function shows the number of bacteria in a pond after t days since the bacteria began to populate the pond. Which of the following is the best interpretation of $(3)^{\frac{t}{14}}$ in the equation?

- ⌗ **A** The number of bacteria at the beginning of the study
<5.0%
-
- ⌗ **B** The number of bacteria triples every two weeks
91.0%
-
- ⌗ **C** The number of bacteria increases by 3 every two weeks
5.9%
-
- ⌗ **D** The number of bacteria in the pond after two weeks
<5.0%

🚩 COME BACK LATER

8.

$$12x^4 - 8x^3 - 15x^2$$

The given expression is equal to $x^2(qx + 5)(rx - 3)$, where $q > 0$ and $r > 0$. What is the value of $q + r$?

PREV. QUESTION

NEXT QUESTION

8

Correct Answer:

8

🚩 COME BACK LATER

9. Line a is parallel to line b . Line a goes through the points $(-4, 0)$ and $(0, 6)$. If line b goes through the point $(-6, 0)$, what is the equation for line b ?

PREV. QUESTION

NEXT QUESTION

⌗ **A** $y = \frac{2}{3}x - 6$
<5.0%

⌗ **B** $y = \frac{2}{3}x + 4$
7.1%

⌗ **C** $y = \frac{3}{2}x + 6$
<5.0%

⌗ **D** $y = \frac{3}{2}x + 9$
85.4%

🚩 COME BACK LATER

10. The exterior angle of an equilateral triangle is equal to $(8x - 20)^\circ$. What is the value of x ?

× ☐ A 5
<5.0%

× ☐ B 10
12.6%

× ☐ C 12.5
6.2%

× ☒ D 17.5
80.0%

PREV. QUESTION

NEXT QUESTION

🚩 COME BACK LATER

11. $(x - 1)(x + 2) = x(x - 3)$

How many solutions does the given equation have?

× ☐ A 0
<5.0%

× ☒ B 1
81.9%

× ☐ C 2
<5.0%

× ☐ D 4
11.1%

PREV. QUESTION

NEXT QUESTION

🚩 COME BACK LATER

12.

$$\begin{aligned}y &> -2x - 1 \\ 3y &< x + 9\end{aligned}$$

Which of the following coordinates would be true for the given system of inequalities?

✕ ☐ A $(-3, 3)$

<5.0%

✕ ☐ B $(-2, 1)$

<5.0%

✕ ☐ C $(1, 4)$

5.0%

✕ ☒ D $(3, 1)$

90.6%

🚩 COME BACK LATER

13.

$$\begin{aligned}9x - 14y &= -3 \\ ax - by &= 6\end{aligned}$$

If the given system of equations has infinite solutions, what is the value of b ?

PREV. QUESTION

NEXT QUESTION

-28

Correct Answer:

-28

🚩 COME BACK LATER

14.

$$f(x) = 6x^2 - 45x + 75$$

Which of the following equivalent forms of the given function f displays the x -intercepts as constants or coefficients?

PREV. QUESTION

NEXT QUESTION

✕ ☒ A $f(x) = (2x - 5)(3x - 15)$

31.8%

✕ ☐ B $f(x) = 3x(2x - 15) + 75$

8.1%

✕ ☐ C $f(x) = 6(x - \frac{15}{4})^2 + \frac{75}{8}$

<5.0%

✕ ☒ D $f(x) = 6(x - 5)(x - \frac{5}{2})$

56.9%

🚩 COME BACK LATER

15.

$$y = 4$$

$$y = x^2 + 12x + n$$

In the given system of equations, n is a positive constant. For what value of n does the system have exactly one real solution?

PREV. QUESTION

NEXT QUESTION

40

Correct Answer:

40

COME BACK LATER

16.

$$y \geq 2(x - 3)^2 + 5$$

$$y \leq x + 3$$

If (a, b) is a solution to the given system of inequalities, what is the minimum value of b ?

PREV. QUESTION

NEXT QUESTION

× A -5
<5.0%

× B -3
<5.0%

× C 3
8.1%

× D 5
85.3%

COME BACK LATER

17. Biology students grew a culture of bacteria at a constant temperature of 55°C for 3 days. The students began with 200 spores and the number of spores doubled every 12 hours. Of the equations given, which can be used to correctly calculate the number of spores in the bacteria culture at the end of 3 days?

PREV. QUESTION

NEXT QUESTION

× A $y = 200(12)^3$
6.0%

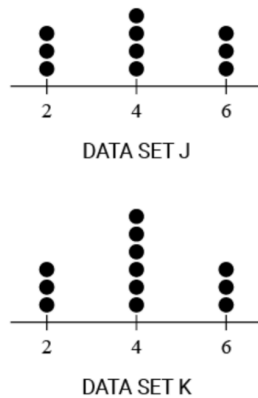
× B $y = 200(2)^3$
8.4%

× C $y = 200(2)^6$
81.7%

× D $y = 200(12)^2$
<5.0%

COME BACK LATER

18.



Which of the statements best compares the means and standard deviations of Data Set J and Data Set K ?

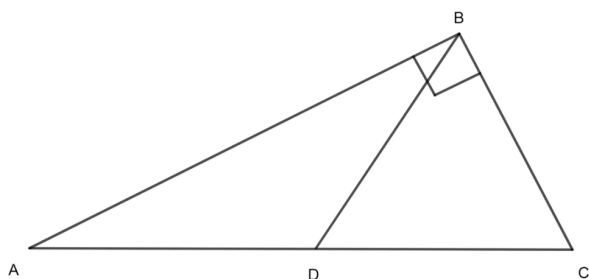
- PREV. QUESTION NEXT QUESTION
- ☒ A The mean of Data Set J is less than the mean of Data Set K , and the standard deviations of the two sets are equal. 19.0%
 - ☐ B The mean of Data Set J is greater than the mean of Data Set K , and the standard deviations of the two sets are equal. 6.0%
 - ☐ C The means of the two sets are equal, and the standard deviation of Data Set J is less than the standard deviation of Data Set K . 27.4%
 - ☒ D The means of the two sets are equal, and the standard deviation of Data Set J is greater than the standard deviation of Data Set K . 47.6%

19. What is the circumference of the circle with an equation of $x^2 + 6x + y^2 - 4y = 51$?

- PREV. QUESTION NEXT QUESTION
- ☐ A 8π <5.0%
 - ☐ B 14.28π 11.9%
 - ☒ C 16π 82.6%
 - ☐ D 64π <5.0%

COME BACK LATER

20.



In the triangle shown, $\angle ABC = 90^\circ$. Line segment BD is shown with point D lying on line AC . What is the value of $\sin \angle ABD - \cos \angle DBC$?

PREV. QUESTION

NEXT QUESTION

☒ A 0
57.1%

☐ B 30
21.0%

☐ C 45
18.1%

☐ D 90
<5.0%

COME BACK LATER

21.

$$y = (x - 6)(x + 6)$$

$$y = k$$

In the system of equations shown, k is an integer constant. What is the largest value of k for which the system has no solution?

PREV. QUESTION

NEXT QUESTION

-37

Correct Answer:

-37

COME BACK LATER

22. The base of the exponential function $g(x)$ is k , where k is a positive real number, and k is equivalent to the value of the y -coordinate of the y -intercept of the graph of $g(x)$. Which of the following functions describes the exponential function $g(x - 1)$?

PREV. QUESTION

NEXT QUESTION

☒ A $g(x - 1) = k^x$
30.0%

☐ B $g(x - 1) = k^{x+1}$
25.6%

☐ C $g(x - 1) = k^x + k$
16.3%

☐ D $g(x - 1) = k^{x-1} + k$
28.1%

COME BACK LATER